FTGP Group 11: Week 22 Sprint Report

EcoPassEU — Privacy-Aware Digital Product Passport DApp

Sprint Duration: 7 April 2026 – 13 April 2026

Report Date: 13 April 2026

University of Bristol

Last Sprint Review and Retrospective

Sprint Duration: 31 March 2026 – 6 April 2026

This section summarises the sprint review meeting held on 6 April 2026 (end of Sprint 0, the initial planning and research sprint). The meeting was attended by Luxiao Cao, Akshansh Rajora, Tianwei Yu, and Detai Dong in person. Fuyu Cao participated online, having been granted leave of absence from the in-person session owing to a mid-term dissertation defence at her undergraduate institution; she nonetheless contributed to all agenda items asynchronously via the group's shared communication channel and reviewed all decisions taken before the close of the meeting.

Completed Work

The following tasks were completed during Sprint 0, together with the primary contributors responsible for each item.

Dataset identification and collection *Luxiao Cao (lead)*
Luxiao Cao took primary responsibility for sourcing and evaluating candidate open datasets suitable for populating the EcoPassEU prototype. Following a comparative review of several databases — including the Global Fashion Agenda's material database, Open Food Facts, and the ESPResso V2 LCA reference tables — the team agreed on *Open Beauty Facts* as the principal dataset for the current sprint. Luxiao Cao downloaded and decompressed the full CSV export (`en.openbeautyfacts.org.products.csv.gz`), performed an initial field audit covering all 76 available columns, and documented the dataset schema, size, and licensing terms (ODbL 1.0 / DbCL 1.0 / CC BY-SA 3.0). He identified that key fields including product name, brand, ingredient lists, packaging descriptions, country of manufacture, and sustainability certifications map directly onto the data model required by ESPR-compliant Digital Product Passports. He also flagged the ESPResso V2 reference tables as a secondary resource — particularly the `base_materials.parquet` emission factors and `aware_factors_agri.csv` water stress characterisation factors — for use in future environmental scoring modules.

Experimental design and methodology *Luxiao Cao, Akshansh Rajora, Tianwei Yu; Fuyu Cao (online)*
Drawing upon the ESPResso V2 LCA framework and ISO 14040/14044 methodology, the team collectively formulated the experimental pipeline for the EcoPassEU prototype. The agreed design proceeds in three stages: (1) ingestion and normalisation of Open Beauty Facts records, including country name standardisation and imputation of missing packaging fields using documented fallback values; (2) serialisation of normalised records to structured JSON-LD documents, off-chain storage on IPFS, and on-chain commitment

of content hashes via an ERC-1155 smart contract on the Polygon Amoy testnet; and (3) functional and non-functional evaluation covering hash integrity verification, QR code resolution, gas cost per DPP issuance, and frontend load time. Fuyu Cao participated in the design discussion online, contributing written comments on the LCA methodology alignment and data gap handling strategy. The methodology was written up and submitted as the group's first weekly progress report in Springer LNCS format.

System architecture design *Akshansh Rajora, Tianwei Yu*
Akshansh Rajora and Tianwei Yu produced the first draft of the system architecture diagram, specifying the interaction between the React frontend, the Ethers.js middleware layer, the Solidity smart contracts, and the IPFS off-chain storage layer. The draft was reviewed by the full group and committed to the repository under `docs/architecture_v1.pdf`.

Smart contract skeleton and development environment *Detai Dong*
Detai Dong initialised the Hardhat development environment and authored a skeleton Solidity smart contract implementing the ERC-1155 semi-fungible token standard. The contract exposes stub functions for `mintBatch`, `updatePassportHash`, and `getPassportURI`, and compiles without errors under Solidity 0.8.24. Unit test scaffolding using Chai and Hardhat's built-in network was also put in place. Detai Dong additionally began planning the React frontend component architecture, identifying the key views required for passport display, QR code resolution, and wallet connection.

Changes and Incomplete Tasks

Data cleaning pipeline — deferred to Sprint 1. The normalisation and cleaning pipeline for the Open Beauty Facts CSV was planned for Sprint 0 but was not completed. The raw file is larger than initially estimated (approximately 2.8 GB decompressed), and the team required additional time to agree on a filtering strategy to produce a manageable and representative working subset. This task has been carried forward as a high-priority item for Sprint 1.

Decentralised Identifier (DID) integration — scope reduced. The original proposal included W3C Decentralised Identifiers (DIDs) and Verifiable Credentials (VCs) as first-class features of the smart contract layer. After architectural review, the team agreed to defer full DID/VC integration to a later sprint and instead simulate the credential structure using JSON-LD documents stored on IPFS. This change reduces scope risk without compromising the conceptual alignment with the ESPR framework.

Fuyu Cao's contributions — adjusted delivery mode. Due to Fuyu Cao's mid-term dissertation defence at her undergraduate institution, her planned contributions to the data pipeline were delivered asynchronously this sprint. She participated in all team discussions online, reviewed all technical decisions, and submitted her written contributions (regulatory notes, dataset evaluation comments) via the group's shared repository. Her role in Sprint 1 will be adjusted to tasks that are better suited to asynchronous collaboration, whilst she remains available online throughout.

Retrospective

Start Doing	Stop Doing	Keep Doing
Set explicit file-size and format constraints when identifying datasets, to avoid late-stage pipeline surprises.	Underestimating the volume of open datasets; always perform a quick field audit before committing to a source.	Daily asynchronous standups via the group chat, which have kept everyone informed without requiring synchronous attendance.
Use feature branches in Git for each sprint task, with pull requests reviewed by at least one other team member before merging.	Leaving architecture decisions undocumented until the review meeting; agree on and record decisions in real time.	Distributing written contributions (notes, drafts) to the shared repository at least 24 hours before meetings to allow async review.
Allocate a dedicated slot in each sprint for integration testing, rather than treating it as an afterthought at the end of development.	Scope creep in technical tasks: DID/VC integration was over-specified for this stage and needed to be deferred.	Inclusive participation: ensuring members attending remotely (as Fuyu Cao did this sprint) have equal access to agenda items and decision records.

Next Sprint Planning

Sprint Duration: 14 April 2026 – 20 April 2026

The sprint planning meeting for Sprint 1 was held on 13 April 2026. The full team was present; Fuyu Cao attended online. The sprint vision, role assignments, and backlog are detailed below.

Product Owner

Akshansh Rajora

Scrum Master

Tianwei Yu

Sprint Vision

By the end of Sprint 1, the team aims to have a working end-to-end prototype connecting the data ingestion pipeline to a functional user-facing interface. A cleaned subset of Open Beauty Facts records should flow through a normalisation script, be committed to IPFS, and have their content hashes stored in the deployed ERC-1155 smart contract on the Polygon Amoy testnet. Crucially, a fully implemented React frontend — covering passport display, QR code resolution, and wallet connection — should be delivered and ready for integration testing by the close of the sprint.

Sprint Backlog

The following tasks have been selected from the product backlog for Sprint 1. Estimates are given in story points (1 point ≈ half a day of focused work).

Task	Description	Assignee(s)	Points
Data cleaning and normalisation	Filter Open Beauty Facts CSV to ∼500 records, resolve encoding issues, standardise country names, and impute missing fields with documented fallback values.	Luxiao Cao (lead), Fuyu Cao (async)	5
Data gap documentation	Produce a structured CSV log of all imputed fields, recording original value, substituted value, and justification.	Fuyu Cao (async)	2
IPFS integration and smart contract	Serialise records to JSON-LD, pin to IPFS, and extend the ERC-1155 contract to store CIDs per token with unit tests.	Tianwei Yu, Akshansh Rajora	6
Testnet deployment	Deploy the smart contract to Polygon Amoy via Hardhat Ignition; commit contract address and ABI to the repository.	Tianwei Yu	2
Frontend: full implementation	Build the complete React frontend: passport display page, QR code scanner/generator, and Ethers.js wallet connection for DPP issuance.	Detai Dong	8
Integration test	End-to-end flow test: ingest → IPFS → mint → QR → display. Record pass/fail, gas costs, and load time.	All members	3

Total story points this sprint: 26

Notes on Member Availability

Fuyu Cao remains on leave of absence from in-person sessions for the duration of Sprint 1 due to her undergraduate mid-term dissertation defence obligations. She will continue to participate online in all planning, review, and retrospective meetings, and will take ownership of the data gap documentation task, which is well-suited to asynchronous delivery. All other team members are available full-time in Bristol for the sprint duration.

Detai Dong has committed to delivering the complete React frontend by the end of Sprint 1. Given the scope of the frontend task (8 story points), this is the highest individual workload in the sprint; the team has agreed to prioritise frontend integration support in the final two days of the sprint to ensure the end-to-end flow test can proceed on schedule.

Anything Else to Share

The team has opened a GitHub repository under a private organisation account for version control and documentation. All sprint artefacts completed to date — including the architecture diagram and dataset schema notes — are committed there. Access has been shared with the supervising teaching staff as requested.

We have also identified a potential risk regarding the IPFS pinning strategy: free-tier pinning services such as Pinata impose monthly bandwidth and storage caps that may become restrictive if the dataset subset grows beyond the planned 500 records. The team will monitor usage during Sprint 1 and evaluate whether a self-hosted IPFS node on a university virtual machine would be more appropriate for the prototype evaluation phase.

Finally, the team notes that the choice of Open Beauty Facts as the primary dataset was confirmed as appropriate during internal review this week. The dataset's broad field coverage (ingredients, packaging, provenance, certifications) maps well onto the DPP data model, and its open licensing allows us to publish example passport records in the repository without restriction. Textile-specific data integration, drawing on the ESPResso V2 carbon and water footprint tables, remains planned for a later sprint.

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